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IEEE Standard for Information Technology— Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks— Specific Requirements

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 6: Light Communications

STANDARDS

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.11bb™-2023

(Amendment to IEEE Std 802.11™-2020
as amended by IEEE Std 802.11ax™-2021,
IEEE Std 802.11ay™-2021,
IEEE Std 802.11ba™-2021,
IEEE Std 802.11az™-2022,
IEEE Std 802.11-2020/Cor 1-2022, and
IEEE Std 802.11bd™-2023)

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Approved 5 June 2023

IEEE SA Standards Board

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Abstract: Modifications to existing physical layers (PHYs) and the medium access control layer (MAC) that enable transparent operation of IEEE 802.11 over the light in the 800 nm to 1000 nm band are specified in this amendment. This amendment specifies PHYs that provide bidirectional operations that achieve a minimum and maximum throughput of 10 Mb/s and 9.6 Gb/s, respectively, as measured at the MAC data service access point (SAP), and facilitate interoperability among solid-state light sources with different modulation bandwidths.

Keywords: IEEE 802.11™, IEEE 802.11bb™, LC, light communication, MAC, medium access control, PHY, physical layer

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Introduction

This introduction is not part of IEEE Std 802.11bb-2023, IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications—Amendment 6: Light Communicatons.

This amendment defines modifications to both the IEEE 802.11 physical layer (PHY) and the IEEE 802.11 medium access control (MAC) sublayer for operation of light communications in the 800 nm to 1000 nm band.

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IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 6: Light Communications

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strikethrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editorial instructions, change markings and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

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2. Normative references

Insert normative reference in alphanumeric order as follows:

IEC 60825-1 Ed. 3.0 b:2014, Safety of laser products—Part 1: Equipment classification and requirements.

3. Definitions, acronyms, and abbreviations

3.1 Definitions

Change the definition as follows:

antenna connector: The measurement point of reference for radio frequency (RF) measurements in a station (STA). The antenna connector is the point in the STA architecture representing the input of the receiver (output of the antenna) for radio reception and the input of the antenna (output of the transmitter) for radio transmission. In systems using multiple antennas or antenna arrays, the antenna connector is a virtual point representing the aggregate output of (or input to) the multiple antennas. In systems using active antenna arrays with processing, the antenna connector is the output of the active array, which includes any processing gain of the active antenna subsystem. In systems using light communications, the antenna connector is a virtual point representing the intermediate frequency (IF) interface of the light communication (LC) optical antenna. In the receive direction, the signal at this virtual point is the output of the LC optical RX antenna and includes any processing gain from the LC optical RX antenna.

Insert the following definition maintaining alphabetical order:

optical front-end (OFE): Optoelectronic module that, in a transmitter, converts an intermediate frequency (IF) electrical signal to an intensity modulated optical signal, and, in a receiver, converts the intensity modulated optical signal to an IF electrical signal.

3.2 Definitions specific to IEEE Std 802.11

Change the definition as follows:

directional multi-gigabit (DMG): Pertaining to operation in a radio frequency (RF) frequency band containing a channel with the channel starting frequency above 45 GHz.

Insert the following definition in alphanumeric order as follows:

light communications station (LC STA): A station (STA) that operates in the light band with wavelengths in the range 800 nm to 1000 nm.

3.4 Acronyms and abbreviations

Insert the following acronym definitions (maintaining alphabetical order):

DC	direct current
IF	intermediate frequency
LC	light communications

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4. General description

4.3 Components of the IEEE Std 802.11 architecture

Insert new subclause 4.3.31 after 4.3.30 EDMG STA, as follows:

4.3.31 Light communication (LC) STA

An LC STA operates in the light band with wavelengths in the range of 800 nm to 1000 nm.

An LC STA has one of three levels of support: HT, VHT, or HE. An LC STA with an LC PHY that has HT support is an HT STA (see 4.3.13) except that it operates in the light band. An LC STA with an LC PHY that has VHT support is a VHT STA (see 4.3.15) except that it operates in the light band. An LC STA with an LC PHY that has HE support is an HE STA (see 4.3.15a) except that it operates in the light band.

An LC STA has support for MAC features such as frame aggregation, block ack, beamforming, and fast session transfer in a multi-band device.

Simultaneous use of multiple transmit chains and multiple receive chains within the LC PHY is supported [see 33.2.3.5 (Multiple transmit chains and multiple receive chains)].

Channels defined for 2.4 GHz operation of HT and HE PHYs do not apply to an LC PHY with HT and HE support, respectively. The channels that apply to LC are defined in Table 33-1.

Insert new Clause 33 after Next Generation V2X (NGV) PHY specification, as follows:

33. Light communication (LC) MAC and PHY specification

33.1 LC MAC specification

33.1.1 Introduction

An LC STA supports the MAC functions defined in Clause 10 (MAC sublayer functional description), the MLME functions defined in Clause 11 (MLME), and the security functions defined in Clause 12 (Security).

33.1.2 LC MAC

An LC STA with HT support shall follow HT MAC procedures and requirements with an LC PHY. An LC STA with VHT support shall follow VHT MAC procedures and requirements with an LC PHY. An LC STA with HE support shall follow HE MAC procedures and requirements with an LC PHY.

33.2 LC PHY specification

33.2.1 General

This subclause specifies the PHY entity for a light communications (LC) orthogonal frequency division multiplexing (OFDM) system with an intensity modulation and direct detection (IM/DD) optical interface.

An LC PHY provides one of three levels of support: HT, VHT, or HE.

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An LC PHY with HT support conforms to the requirements for an HT PHY as defined in Clause 19 except where the requirements in 33.2 supersede those requirements.

An LC PHY with VHT support conforms to the requirements for a VHT PHY as defined in Clause 21 except where the requirements in 33.2 supersede those requirements.

An LC PHY with HE support conforms to the requirements for an HE PHY as defined in Clause 27 except for the requirements in 33.2 that supersede those requirements.

33.2.2 LC PHY service interface

The LC PHY service is provided to the MAC through the PHY service primitive described in Clause 8 (PHY service specification).

33.2.3 LC PHY

33.2.3.1 Introduction

The procedure by which an LC PHY converts a PSDU into a PPDU is defined in 19.3 (HT PHY), 21.3 (VHT PHY), and 27.3 (HE PHY) for an LC PHY with HT, VHT and HE support, respectively, except for the differences described in the remainder of 33.2.3 (LC PHY).

For the requirements in 21.3 (VHT PHY) and 27.3 (HE PHY), the term *frequency segment* applies to the LC IF signal.

33.2.3.2 Transmitter block diagram

The transmitter block diagrams in 19.3.3 (Transmitter block diagram), 21.3.3 (Transmitter block diagram), and 27.3.5 (Transmitter block diagram) show an Analog and RF block that is described as upconverting the complex baseband waveform associated with each transmit chain to an RF signal.

In the LC PHY, the complex baseband waveform associated with each transmit chain is instead upconverted to an LC IF signal, either directly or by up/down conversion. A DC bias is added to the LC IF signal which is then fed into an optical front end (OFE). The OFE converts the DC biased LC IF signal into an intensity modulated optical signal.

In comparison to an RF channel, which is characterized by multipath propagation, the LC channel is relatively transparent with a dominant LOS component usually 10s of decibels stronger than non-line-of-sight components.

In some implementations, the complex baseband waveform is upconverted directly to an LC IF signal as shown in Figure 33-1 (Direct conversion LC TX). In this example, the LC PHY TX/RX are HT/VHT/HE PHY TX/RX where the RF antennas are replaced by LC optical antennas.

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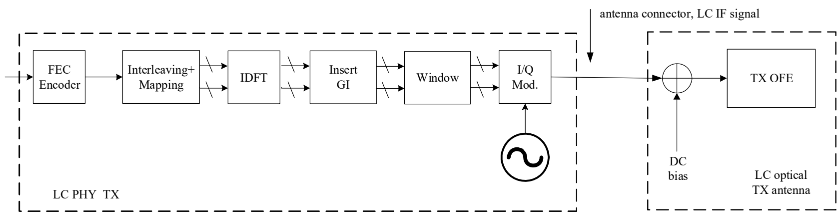

Figure 33-1—Direct conversion LC TX

In some implementations, the complex baseband waveform might first be upconverted to an RF signal and then downconverted to the LC IF signal as shown in Figure 33-2 (Up/down conversion LC TX). The downconversion is defined to have unity gain.

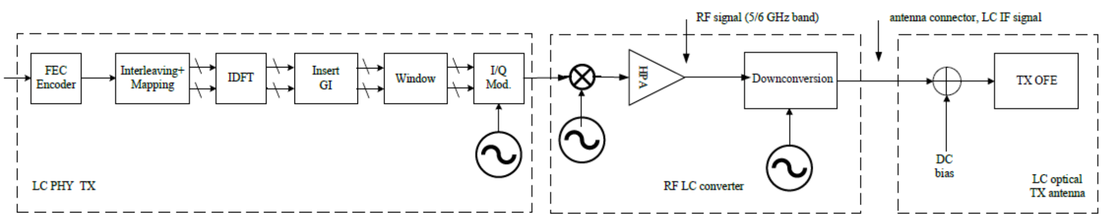

Figure 33-2—Up/down conversion LC TX

In this example, the RF signal from the HPA, which is in the frequency range 5 GHz to 6 GHz, is downconverted such that the center frequency aligns with the LC IF channel frequency defined in 33.2.3.4 (Channel numbering). The reference clock of the local oscillator for the downconversion at the LC optical TX antenna is the same as in the LC PHY TX.

A DC bias is added to the LC IF signal before the signal is fed to the transmitting OFE because the current through an SSL device can only be positive, as illustrated in Figure 33-3 (Operation of an SSL device with DC bias).

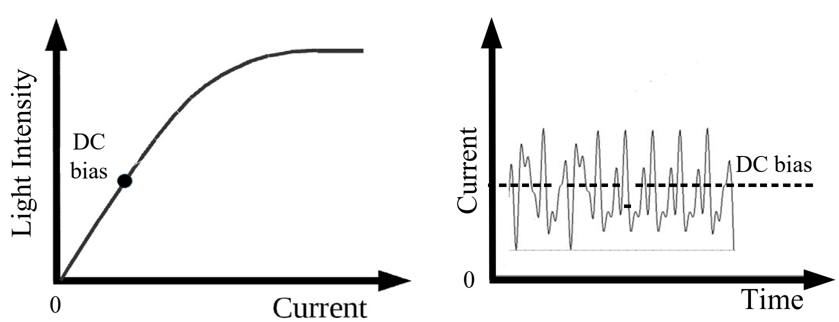

Figure 33-3—Operation of an SSL device with DC bias

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An LC PHY supports the use of multiple transmit chains and multiple receive chains. An example of the LC PHY TX connected to multiple LC optical TX antennas is shown in Figure 33-4 (Connecting the LC PHY TX to multiple optical TX antennas).

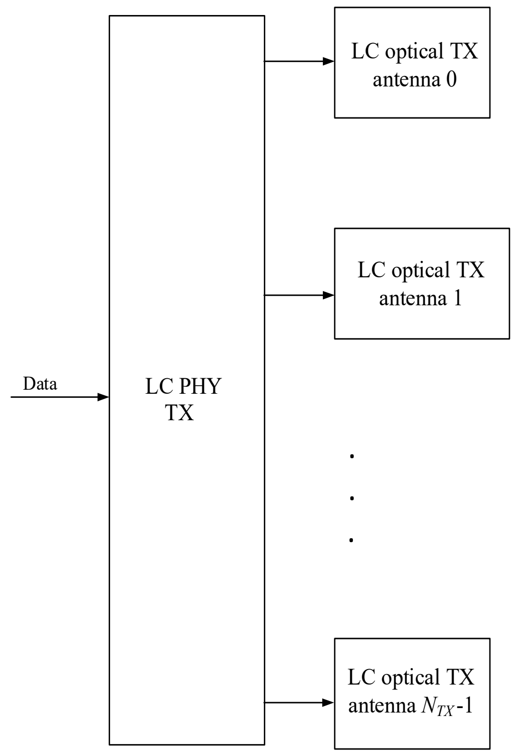

Figure 33-4—Connecting the LC PHY TX to multiple optical TX antennas

33.2.3.3 Mathematical description of signals

The LC IF signal is described by Equation (19-1), Equation (21-11), and Equation (27-1) for an LC PHY with HT, VHT or HE support, respectively, where f_c is the LC IF.

33.2.3.4 Channel numbering

In systems using light communications, the frequency segment refers to the LC IF signal.

LC IF channel center frequencies are defined at integer multiples of 5 MHz above the LC IF channel starting frequency. The relationship between LC IF channel center frequency and channel number n_{ch} is given in Equation (33-1).

$$\text{LC IF channel center frequency} = \text{LC IF channel starting frequency} + 5 \times n_{ch}\ (\text{MHz}), \tag{33-1}$$

where

$n_{ch} = 1,\ldots, 64$ is the channel number or channel center frequency index

LC IF channel starting is defined in Annex E

Equation (33-1) and the LC IF channel starting frequencies defined in Annex E divide the LC IF spectrum into two segments: frequencies in the range 16 MHz to 176 MHz that map from the 5 GHz band and

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frequencies in the range 176 MHz to 336 MHz that map from the 6 GHz band as shown in Table 33-1 (RF to LC IF mapping for channels in the 5 GHz and 6 GHz bands).

Table 33-1—RF to LC IF mapping for channels in the 5 GHz and 6 GHz bands

				Channel width					
Channel number	**RF frequency band**	**RF center frequency (MHz)**	**LC IF center frequency (MHz)**	**20 MHz**	**40 MHz PrimaryChannel LowerBehavior**	**40 MHz PrimaryChannel UpperBehavior**	**40 MHz**	**80 MHz / 80+80 MHz**	**160 MHz**
34	5 GHz	5170	16	Channel number 36 16 MHz–36 MHz	Channel number 36 16 MHz–56 MHz	Channel number 40 16 MHz–56 MHz	NA	Channel center frequency index 42 16 MHz–96 MHz	Channel center frequency index 50 16 MHz–176 MHz
36		5180	26						
38		5190	36	Channel number 40 36 MHz–56 MHz					
40		5200	46						
42		5210	56	Channel number 44 56 MHz–76 MHz	Channel number 44 56 MHz–96 MHz	Channel number 48 56 MHz–96 MHz			
44		5220	66						
46		5230	76	Channel number 48 76 MHz–96 MHz					
48		5240	86						
50		5250	96	Channel number 52 96 MHz–106 MHz	Channel number 52 96 MHz–136 MHz	Channel number 56 96 MHz–136 MHz		Channel center frequency index 58 96 MHz–176 MHz	
52		5260	106						
54		5270	116	Channel number 56 116 MHz–136 MHz					
56		5280	126						
58		5290	136	Channel number 60 136 MHz–156 MHz	Channel number 60 136 MHz–176 MHz	Channel number 64 136 MHz–176 MHz			
60		5300	146						
62		5310	156	Channel number 64 156 MHz–176 MHz					
64		5320	166						
	6 GHz		176	Channel number 1 176 MHz–196 MHz	NA	NA	Channel center frequency index 3 176 MHz–216 MHz	Channel center frequency index 7 176 MHz–256 MHz	Channel center frequency index 15 176 MHz–336 MHz
1		5955	186						
3		5965	196	Channel number 5 196 MHz–216 MHz					
5		5975	206						
7		5985	216	Channel number 9 216 MHz–236 MHz			Channel center frequency index 11 216 MHz–256 MHz		
9		5995	226						
11		6005	236	Channel number 13 236 MHz-256 MHz					
13		6015	246						
15		6025	256	Channel number 17 256 MHz–276 MHz			Channel center frequency index 19 256 MHz–296 MHz	Channel center frequency index 23 256 MHz–336 MHz	
17		6035	266						
19		6045	276	Channel number 21 276 MHz–296 MHz					
21		6055	286						
23		6065	296	Channel number 25 296 MHz–316 MHz			Channel center frequency index 27 296 MHz–336 MHz		
25		6075	306						
27		6085	316	Channel number 29 316 MHz–336 MHz					
29		6095	326						

A channel number or channel center frequency index from the 5 GHz band shall not be greater than 64. A channel number or center frequency index from the 6 GHz band shall not be greater than 29.

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33.2.3.5 Multiple transmit chains and multiple receive chains

33.2.3.5.1 General

The LC PHY may use multiple transmit chains and multiple receive chains to support spatial multiplexing and/or wavelength division multiplexing.

In an LC PHY that implements RF up/down conversion and has multiple transmit chains, the RF downconverters shall be synchronized to a common reference clock. In an LC PHY that implements RF up/downconversion and has multiple receive chains, the RF upconverters shall be synchronized to a common reference clock.

For an LC PHY with multiple optical TX antennas and HT support, the LC PHY TX shall use the procedures defined in 19.3 (HT PHY), where N_{TX} transmit chains in the HT PHY shall be connected to N_{TX} LC optical TX antennas.

For an LC PHY with multiple optical TX antennas and VHT support, the LC PHY TX shall use the procedures defined in 21.3 (VHT PHY), where N_{TX} transmit chains in the VHT PHY shall be connected to N_{TX} LC optical TX antennas.

For an LC PHY with multiple optical TX antennas and HE support, the LC PHY TX shall use the procedures defined in 27.3 (HE PHY), where N_{TX} transmit chains in the HE PHY shall be connected to N_{TX} LC optical TX antennas.

33.2.3.5.2 Spatial muliplexing

Spatial multiplexing is supported when LC optical RX antennas and LC optical TX antennas are positioned such that light transmitted by an LC optical TX antenna incident on an LC optical RX antenna is at least partially isolated from the light transmitted by another LC optical TX antenna that is incident on another LC optical RX antenna. The isolation might be achieved by directing the light at the optical TX antenna or capturing light from a particular direction at the optical RX antenna. The isolation might also be achieved by spatially separating the LC optical TX antennas and/or LC optical RX antennas. This principle is illustrated in Figure 33-5 (Spatial multiplexing with an LC PHY).

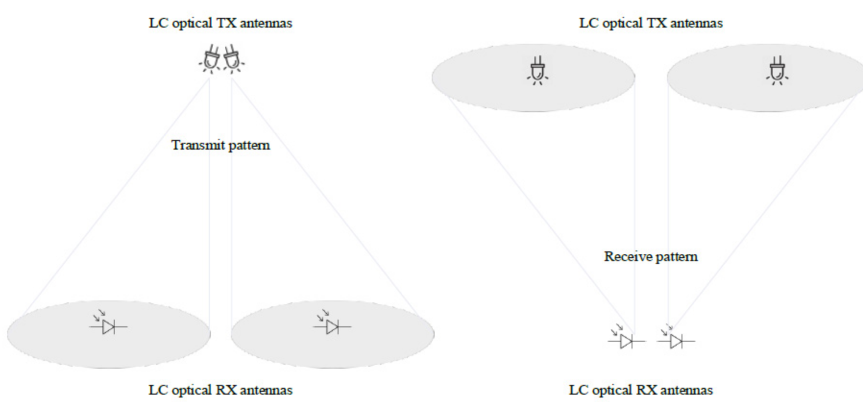

Figure 33-5—Spatial multiplexing with an LC PHY

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33.2.3.5.3 Wavelength-division multiplexing

Wavelength division multiplexing is supported when the wavelength of the light transmitted by one LC optical TX antenna is different from the wavelength of the light transmitted by another LC optical TX antenna and, correspondingly, one LC optical RX antenna is sensitive to the light of the first wavelength but not the second wavelength and another LC optical RX antenna is sensitive to the light of the second wavelength but not the first wavelength. This principle is illustrated in Figure 33-6 (Wavelength division multiplexing with an LC PHY).

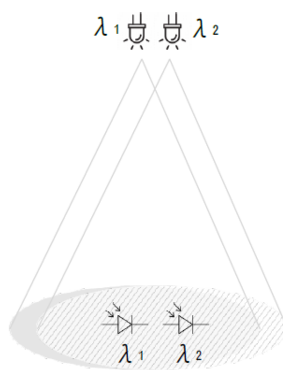

Figure 33-6—Wavelength division multiplexing with an LC PHY

For example, with two transmit chains, one operating at 950 nm and the other at 850 nm, a shortpass dicroic beam splitter with a cut-off wavelength of 900 nm can be used in front of both receive chains. The wavelength of the first transmit chain is longer than the cut-off wavelength. It is reflected and detected by the first receive chain. The wavelength of the second transmit chain is shorter than the cut-off wavelength. It passes through and is detected by the second receive chain.

An LC STA with a single transmit chain is interoperable with an LC STA with multiple transmit and multiple receive chains operating in wavelength division multiplexing mode. The sum of transmitted and reflected signals from a dicroic beamsplitter is approximately the same, independent of the wavelength.

For example, a first LC STA with a single optical antenna supports a single stream. This stream can be detected by both receive chains of a second LC STA with two optical antennas by combining their signals. In the reverse link direction, both transmit chains of the second LC STA are used to transmit a single stream which can be detected by the single receive chain of the first LC STA.

Note that when using LEDs with different wavelengths, their optical spectra might be wider and cause residual crosstalk at a receive chain. The crosstalk might be minimized by using multiple-transmit multiple-receive antenna processing.

In the above description, if the LC optical TX antennas are in one LC STA and the LC optical RX antennas are in another LC STA the arrangement supports SU-MIMO. If the LC optical antennas (TX or RX) at one end are in the same LC STA but the LC optical antennas (RX or TX) at the other end are in different LC STAs then the arrangement supports MU-MIMO.

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An LC optical RX antenna shall be sensitive to light in one of the following wavelength ranges:

- 800 nm to 900 nm
- 900 nm to 1000 nm
- 800 nm to 1000 nm

An LC PHY shall have at least one receive chain with an LC optical RX antenna sensitive to light in the 800 nm to 1000 nm range or at least two receive chains with one receive chain with an LC optical RX antenna sensitive to light in the 800 nm to 900 nm range and the other receive chain with an LC optical RX antenna sensitive to light in the 900 nm to 1000 nm range.

The maximum number of spatial streams supported by an LC PHY using wavelength division multiplexing is 2.

33.2.3.6 Receive specification

The minimum receive sensitivity of an LC PHY shall be –15 dBm_{opt} measured as the average incident optical power in the range 800 nm to 1000 nm at the LC optical RX antennas.

33.2.3.7 CCA for LC

33.2.3.7.1 CCA requirements

The CCA requirements for the LC PHY apply to the LC IF signal as measured at the antenna connector.

The CCA requirements for the HT PHY in 19.3.19.5 (CCA sensitivity) to detect a channel busy condition are applicable to the LC PHY with HT support.

The CCA requirements for the VHT PHY in 21.3.18.5 (CCA sensitivity) to detect a channel busy condition are applicable to the LC PHY with VHT support.

The CCA requirements for the HE PHY in 27.2.6 (Support for non-HT, HT and VHT formats) and 27.3.20 (Receiver specification) to detect a channel busy condition are applicable to LC PHY with HE support.

33.2.3.7.2 LC repetition

An LC STA transmission is directional, hence a non-AP LC STA might not be able to receive signals from peer non-AP LC STAs. LC repetition is an optional feature in an LC AP that effectively extends the area covered by a transmission from a non-AP LC STA to include the area covered by the LC AP. LC repetition together with other mechanisms, such as the RTS/CTS mechanism defined in 10.3 (DCF), might reduce collisions in an LC BSS.

An LC AP may transmit at the TX OFE a repeat of the signal it receives at the RX OFE. The delay between the received signal and the repeated signal shall be less than 1 µs.

Figure 33-7 (An example of channel access with LC repetition) illustrates an example of channel access with the LC repetition mechanism.

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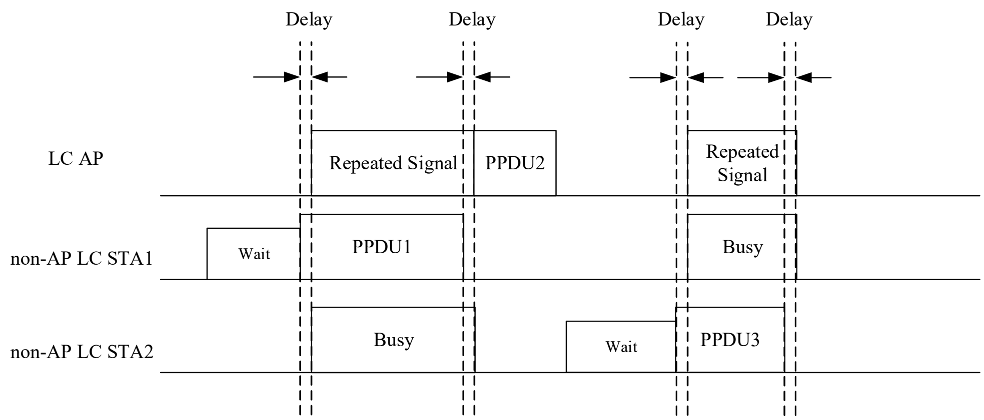

Figure 33-7—An example of channel access with LC repetition

The LC AP receives PPDU1 from non-AP LC STA1 and repeats the signal to non-AP LC STAs in its coverage area. The non-AP LC STA2 then detects a channel busy condition on receiving the repeated signal. After retransmitting PPDU1, the LC AP sends PPDU2 to non-AP LC STA1 in response to PPDU1. Non-AP LC STA2 sends PPDU3 to the LC AP after it detects an idle channel condition. The LC AP repeats the signal and, as a result, non-AP LC STA1 finds the channel busy.

33.2.3.8 Regulatory requirements

Wireless LANs (WLANs) implemented in accordance with this standard are subject to equipment certification and operating requirements established by regional and national regulatory administrations. The PHY specification establishes minimum technical requirements for interoperability, based upon established regulations at the time this standard was issued. These regulations are subject to revision or may be superseded. Requirements that are subject to local geographic regulations are annotated within the PHY specification. Regulatory requirements that do not affect interoperability are not addressed in this standard.

Implementers are referred to the regulatory sources in Annex D and to IEC 60825-1 (or its relevant latest version) for further information.

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Annex B

(normative)

Protocol Implementation Conformance Statement (PICS) proforma

B.2.2 General abbreviations for item and Support columns

Insert the following acronym:

LC light communications

B.4 PICS proforma—IEEE Std 802.11-2020

Insert the following row in B.4.3 as follows (note that the entire table is not shown):

B.4.3 IUT configuration

Item	IUT configuration	References	Status	Support
* CFLC	light communications	—	4.3.31 [Light communications (LC) STA]	Yes ☐ No ☐ N/A ☐

Insert new subclause after B.4.37 as follows:

B.4.38 Light communications (LC) features

Item	Protocol capability	References	Status	Support
	Are the following PHY protocol features supported?			
LCP1	LC PHY level of support			
*LCP1.1	LC PHY with HT support	33.2.1 General	CFHT5G: O.1	Yes □ No □ N/A □
*LCP1.2	LC PHY with VHT support	33.2.1 General	CFVHT: O.1	Yes □ No □ N/A □
*LCP1.3	LC PHY with HE support	33.2.1 General	CFHE5G OR CFHE6G: O.1	Yes □ No □ N/A □
LCP3	Channel numbering and channelization	33.2.3.4 (Channel numbering)		
LCP3.1	5 GHz band mapping	33.2.3.4 (Channel numbering)	CFLC: O.2	Yes □ No □ N/A □

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LCP3.2	6 GHz band mapping	33.2.3.4 (Channel numbering)	CFLC: O.2	Yes □ No □ N/A □
LCP4	Multiple transmit and receive chains	33.2.3.5 (Multiple transmit chains and multiple receive chains)		
LCP4.1	More than one transmit chain	33.2.3.5 (Multiple transmit chains and multiple receive chains)	CFLC: O	Yes □ No □ N/A □
LCP4.2	More than one receive chain	33.2.3.5 (Multiple transmit chains and multiple receive chains)	CFLC: O	Yes □ No □ N/A □
LCP5	LC CCA requirements	33.2.3.7.1 (CCA requirements)	CFLC: M	Yes □ No □ N/A □
LCP6	LC repetition mechanism	33.2.3.7.2 (LC repetition)	CFLC: O	Yes □ No □ N/A □

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Annex C

(normative)

ASN.1 encoding of the MAC and PHY MIB

C.3 MIB Detail

Change dot11StationConfigEntry as follows:

```
Dot11StationConfigEntry ::= SEQUENCE
{
 …
dot11StationMeasurementPeriod Unsigned32,
dot11LCOptionImplemented TruthValue
```

Insert the following dot11StationMeasurementPeriod object definition:

```
  (llbb)dot11LCOptionImplemented OBJECT-TYPE
  SYNTAX TruthValue
  MAX-ACCESS read-only
  STATUS current
  DESCRIPTION
  "This is a capability variable.
  Its value is determined by device capabilities.
  This attribute indicates whether the entity is LC capable."

::= { dot11StationConfigEntry 226 }
```


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Annex E

(normative)

Country element and operating classes

E.1 Country information and operating classes

Change Table E-1 as follows (note that the entire table is not shown):

Table E-1—Operating classes in the United States

Operating class	Nonglobal operating class	Channel starting frequency (GHz)	Channel spacing (MHz)	LC IF Channel starting frequency (MHz)	Channel set	Channel center frequency index	Behavior limits set
1	115	5	20	–154	36, 40, 44, 48	—	Applicable to LC
2	118	5	20	–154	52, 56, 60, 64	—	DFS_50_100_Behavior, Applicable to LC
...	...	...	...	...	...	...	...
22	116	5	40	–154	36, 44	—	PrimaryChannelLowerBehavior, Applicable to LC
23	119	5	40	–154	52, 60	—	PrimaryChannelLowerBehavior, Applicable to LC
...	...	...	...	...	...	...	...
27	117	5	40	–154	40, 48	—	PrimaryChannelUpperBehavior, Applicable to LC
28	120	5	40	–154	56, 64	—	PrimaryChannelUpperBehavior, Applicable to LC
...	...	...	...	...	...	...	...
128	128	5	80	–154	—	42, 58, 106, 122, 138, 155	UseEirpForVHTTxPowEnv, Applicable to LC
129	129	5	160	–154	—	50, 114	UseEirpForVHTTxPowEnv, Applicable to LC

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Change Table E-2 as follows (note entire table not shown):

Table E-2—Operating classes in Europe

Operating class	Global operating class (see Table 4)	Channel starting frequency (GHz)	Channel spacing (MHz)	LC IF Channel starting frequency (MHz)	Channel set	Channel center frequency index	Behavior limits set
1	115	5	20	–154	36, 40, 44, 48	—	Applicable to LC
2	118	5	20	–154	52, 56, 60, 64	—	NomadicBehavior Applicable to LC
...	...	...	...	...	...	...	...
5	116	5	40	–154	36, 44	—	PrimaryChannelLowerBehavior, Applicable to LC
6	119	5	40	–154	52, 60	—	PrimaryChannelLowerBehavior, Applicable to LC
...	...	...	...	...	...	...	...
8	117	5	40	–154	40, 48	—	PrimaryChannelUpperBehavior, Applicable to LC
9	120	5	40	–154	56, 64	—	PrimaryChannelUpperBehavior, Applicable to LC
...	...	...	...	...	...	...	...
128	128	5	80	–154	—	42, 58, 106, 122	UseEirpForVHTTxPowEnv, Applicable to LC
129	129	5	160	–154	—	50, 114	UseEirpForVHTTxPowEnv, Applicable to LC
130	130	5	80	–154	—	42, 58, 106, 122	80+, UseEirpForVHTTxPowEnv, Applicable to LC

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Change Table E-3 as follows (note entire table not shown):

Table E-3—Operating classes in Japan

Operating class	Global operating class (see Table 4)	Channel starting frequency (GHz)	Channel spacing (MHz)	LC IF Channel starting frequency (MHz)	Channel set	Channel center frequency index	Behavior limits set
1	115	5	20	–154	34, 38, 42, 46[a] 36, 40, 44, 48	—	Applicable to LC
...	...	...	...		...	...	...
32	118	5	20	–154	52, 56, 60, 64	—	Applicable to LC
33	118	5	20	–154	52, 56, 60, 64	—	Applicable to LC
...	...	...	...		...	...	...
36	116	5	40	–154	36, 44	—	PrimaryChannelLowerBehavior, Applicable to LC
37	119	5	40	–154	52, 60	—	PrimaryChannelLowerBehavior, Applicable to LC
...	...	...	...		...	...	...
41	117	5	40	–154	40, 48	—	PrimaryChannelUpperBehavior, Applicable to LC
42	120	5	40	–154	56, 64	—	PrimaryChannelUpperBehavior, Applicable to LC
...	...	...	...	...	...	...	...
128	128	5	80	–154	—	42, 58, 106, 122, 138	UseEirpForVHTTxPowEnv, Applicable to LC
129	129	5	160	–154	—	50, 114	UseEirpForVHTTxPowEnv, Applicable to LC
130	130	5	80	–154	—	42, 58, 106, 122, 138	80+, UseEirpForVHTTxPowEnv, Applicable to LC

[a] The use of channels 34, 38, 42, and 46 was included in the initial 5 GHz rules in May 2005 and cannot be used after May 2012.

Change Table E-4 as follows (note entire table not shown):

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Table E-4—Global operating classes

Operating class	Nonglobal operating class(es) (see NOTE 3)	Channel starting frequency (GHz)	Channel spacing (MHz)	LC IF Channel starting frequency (MHz)	Channel set	Channel center frequency index	Behavior limits set
115	1-1, 2-1, 3-1, E-6-1	5	20	–154	36, 40, 44, 48	—	UseEirpForVHTTxPowEnv, Applicable to LC
116	1-22, 2-5, 3-36, E-6-4	5	40	–154	36, 44	—	PrimaryChannelLowerBehavior, UseEirpForVHTTxPowEnv, Applicable to LC
117	1-27, 2-8, 3-41	5	40	–154	40, 48	—	PrimaryChannelUpperBehavior, UseEirpForVHTTxPowEnv, Applicable to LC
118	1-2, 2-2, 3-32,33, E-6-2	5	20	–154	52, 56, 60, 64	—	DFS_50_100_Behavior, UseEirpForVHTTxPowEnv, Applicable to LC
119	1-23, 2-6, 3-37,38, E-6-5	5	40	–154	52, 60	—	PrimaryChannelLowerBehavior, DFS_50_100_Behavior, UseEirpForVHTTxPowEnv, Applicable to LC
120	1-28, 2-9, 3-42,43	5	40	–154	56, 64	—	PrimaryChannelUpperBehavior, DFS_50_100_Behavior, UseEirpForVHTTxPowEnv, Applicable to LC
...	...	...	...		...	...	...
128	1-128, 2-128, 3-128, E-6-128	5	80	–154	—	42, 58, 106, 122, 138, 155	UseEirpForVHTTxPowEnv, Applicable to LC
129	1-129, 2-129, 3-129, E-6-129	5	160	–154	—	50, 114	UseEirpForVHTTxPowEnv, Applicable to LC
130	1-130, 2-130, 3-130, E-6-130	5	80	–154	—	42, 58, 106, 122, 138, 155	80+, UseEirpForVHTTxPowEnv, Applicable to LC

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Table E-4—Global operating classes *(continued)*

Operating class	Nonglobal operating class(es) (see NOTE 3)	Channel starting frequency (GHz)	Channel spacing (MHz)	LC IF Channel starting frequency (MHz)	Channel set	Channel center frequency index	Behavior limits set
131		5.950	20	–181	1, 5, 9, 13, 17, 21, 25, 29, 33, 37, 41, 45, 49, 53, 57, 61, 65, 69, 73, 77, 81, 85, 89, 93, 97, 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, 141, 145, 149, 153, 157, 161, 165, 169, 173, 177, 181, 185, 189, 193, 197, 201, 205, 209, 213, 217, 221, 225, 229, 233	—	Applicable to LC
132		5.950	40	–181	—	3, 11, 19, 27, 35, 43, 51, 59, 67, 75, 83, 91, 99, 107, 115, 123, 131, 139, 147, 155, 163, 171, 179, 187, 195, 203, 211, 219, 227	Applicable to LC

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Table E-4—Global operating classes *(continued)*

Operating class	**Nonglobal operating class(es)** (see NOTE 3)	**Channel starting frequency (GHz)**	**Channel spacing (MHz)**	**LC IF Channel starting frequency (MHz)**	**Channel set**	**Channel center frequency index**	**Behavior limits set**
133		5.950	80	–181	—	7, 23, 39, 55, 71, 87, 103, 119, 135, 151, 167, 183, 199, 215	Applicable to LC
134		5.950	160	–181	—	15, 47, 79, 111, 143, 175, 207	Applicable to LC
135		5.950	80	–181	—	7, 23, 39, 55, 71, 87, 103, 119, 135, 151, 167, 183, 199, 215	80+ Applicable to LC

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